

Research trend mini-review

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Trend: Testing AI systems

Metamorphic Testing: A Review of Challenges and Opportunities (academic source)

This article deals with the **test oracle problem**. This problem occurs when it is challenging, impossible, or too expensive to check if the output of a single test is correct. Examples include finding the shortest path in a large graph, compilers, machine-learning programs, and search engines.

The authors present metamorphic testing (MT) as a way to both create new tests and check test results. Instead of checking each output against a known correct answer, MT checks if the outputs of several related tests follow certain rules. These rules are called metamorphic relations (MRs). For example, if you swap the start and end points in a shortest-path search, the length of the path should remain unchanged. If it does not, the program must contain a fault.

The article shows that MT works well in practice. It has found real bugs in well-tested software, including over 100 bugs in the GCC and LLVM compilers. One study mentioned in the article found that only three to six different MRs were enough to find at least 90% of the faults that a full test oracle would find. However, the authors point out that MT reduces the oracle problem but does not fully solve it. The article also explains the key ideas behind MT, clears up common misunderstandings, and lists open challenges, such as how to find and choose good MRs.

The ML Test Score: A Rubric for ML Production Readiness and Technical Debt Reduction (practitioner source)

While traditional testing often focuses on failures within the codebase, Google shows that failures in ML-systems often are caused by data, infrastructure, and silent failures (where the system does not crash, but gives bad answers).

The authors divide testing of ML systems into four concrete areas:

- **Features & Data:** Schema validation, data invariants, and the cost of input features.
- **Model Development:** Testing against simple baseline models, fairness/inclusion, and model staleness.
- **Infrastructure:** Deterministic training, fast unit tests for model specifications, integration tests, and safe rollbacks/canary deploys.
- **Monitoring:** Monitoring of training/serving skew and numerical stability in production.

Justification of test improvement

Given that failures regarding ML-systems often are caused by data - metamorphic testing is a great tool for testing data invariants, such as **irrelevant details** and **order invariance** which we tested for in activity 5.3.

Sources

- [Metamorphic testing: A review of challenges and opportunities](#) (academic source)
- [The ML Test Score: A rubric for ML production systems](#) (practitioner source)